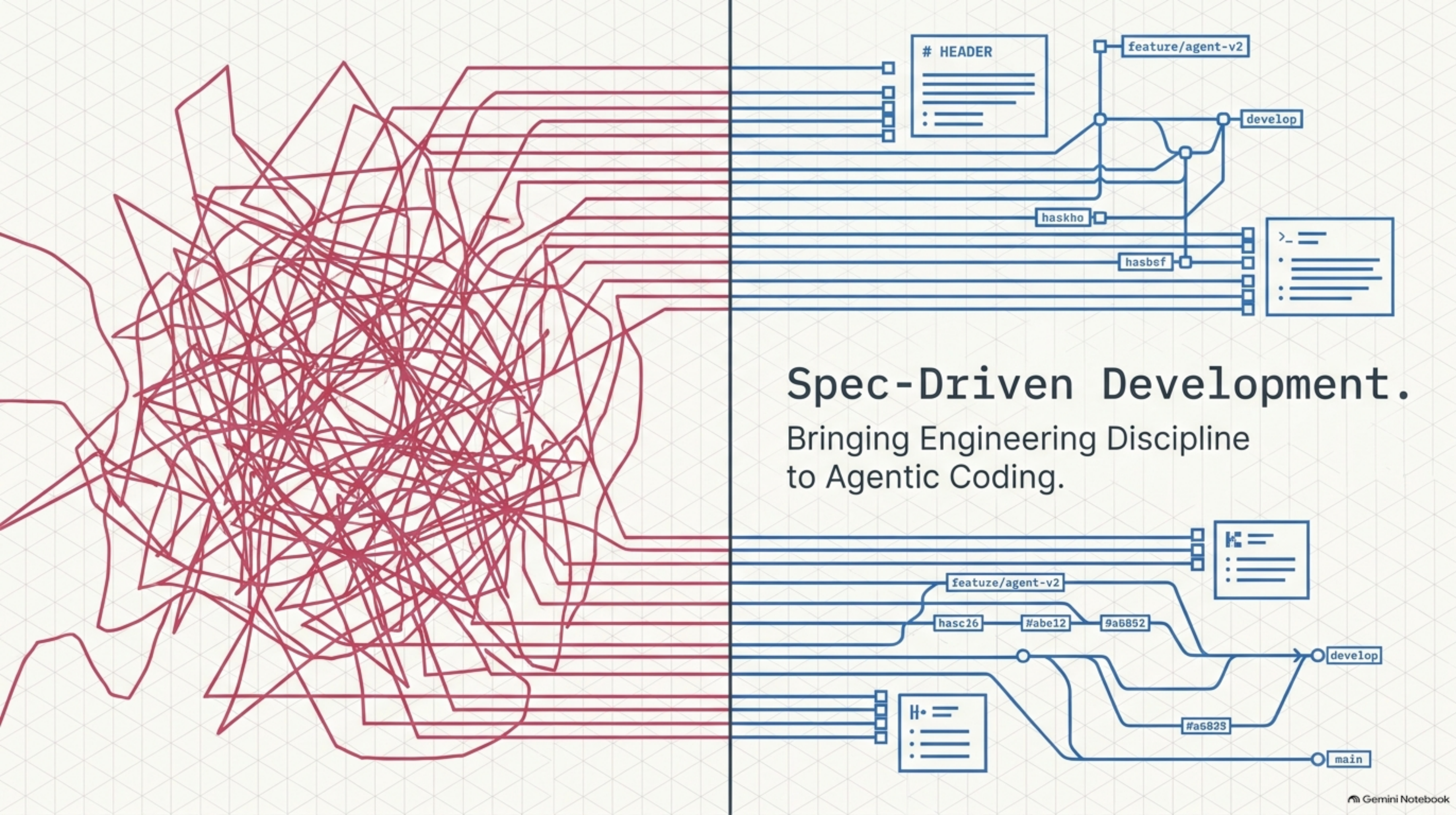
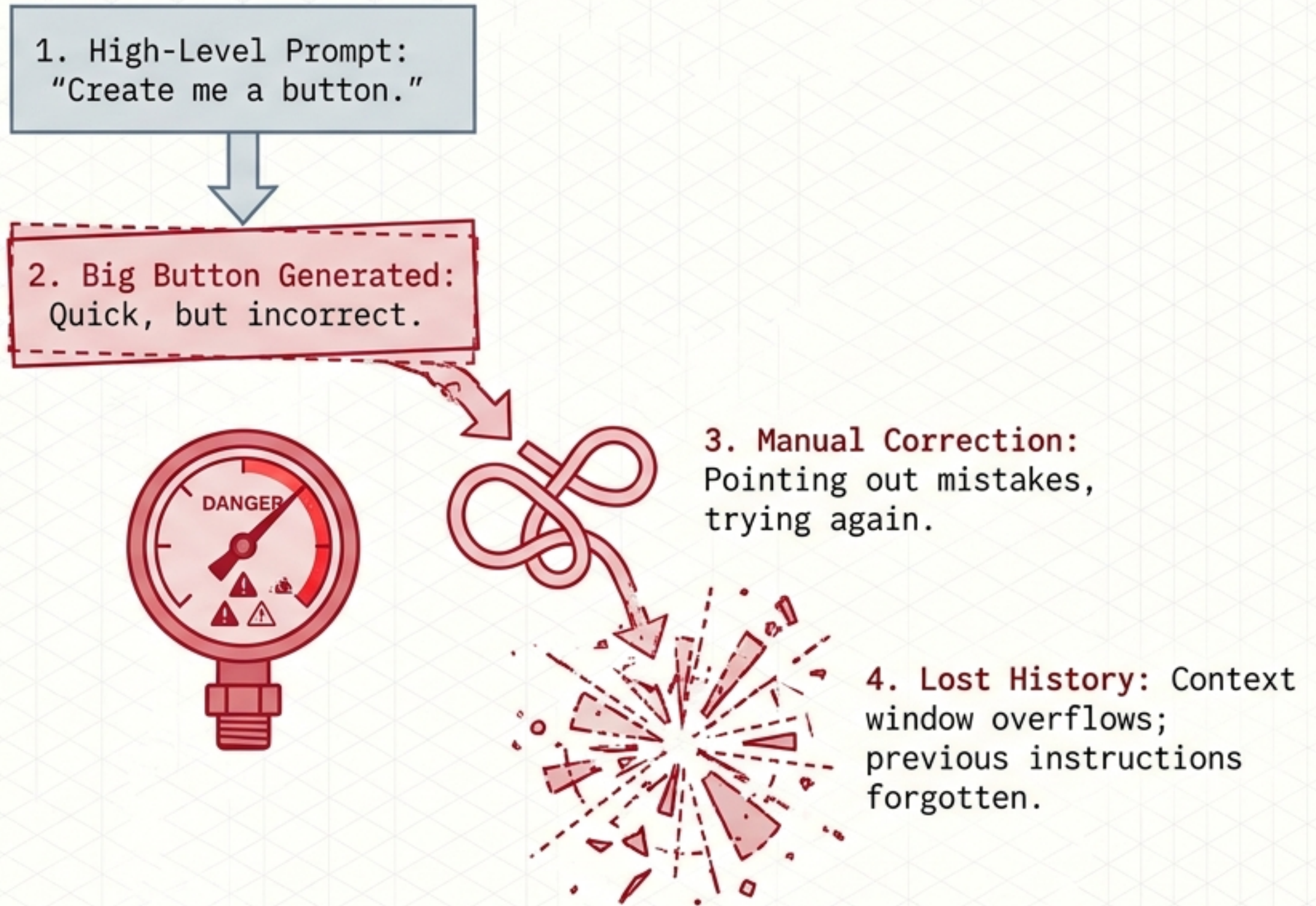




The Chaos of Vibe Coding



Vibe coding is the chaos of unsupervised AI generation. It works for a button, but fails at scale. The result is mounting technical debt and severe AI fatigue.

The Paradigm Shift

The Brain/Architect



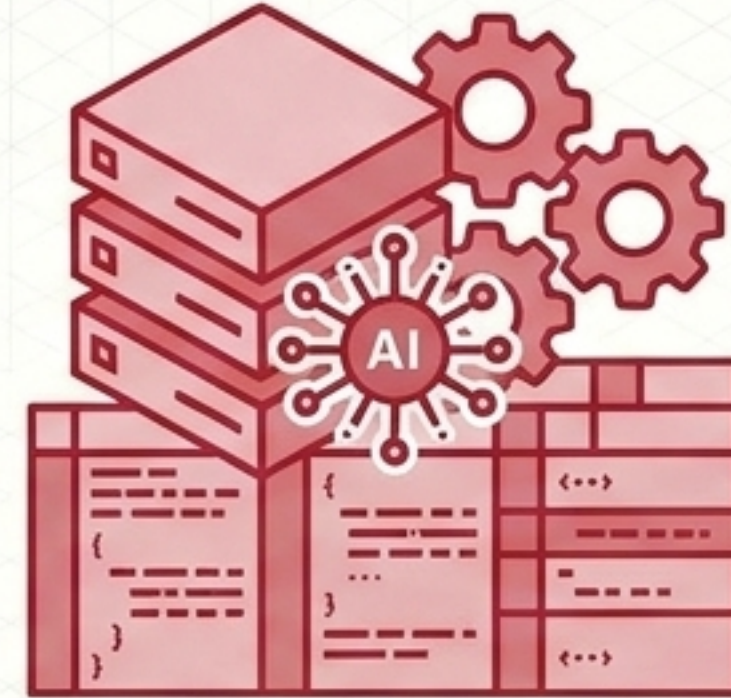
Defines the What & Why

Generates the Specification
(Permanent Intent)



The
Contract Line

The Muscle/Builder



Executes the How



Generates the **Implementation**
(Disposable Code)

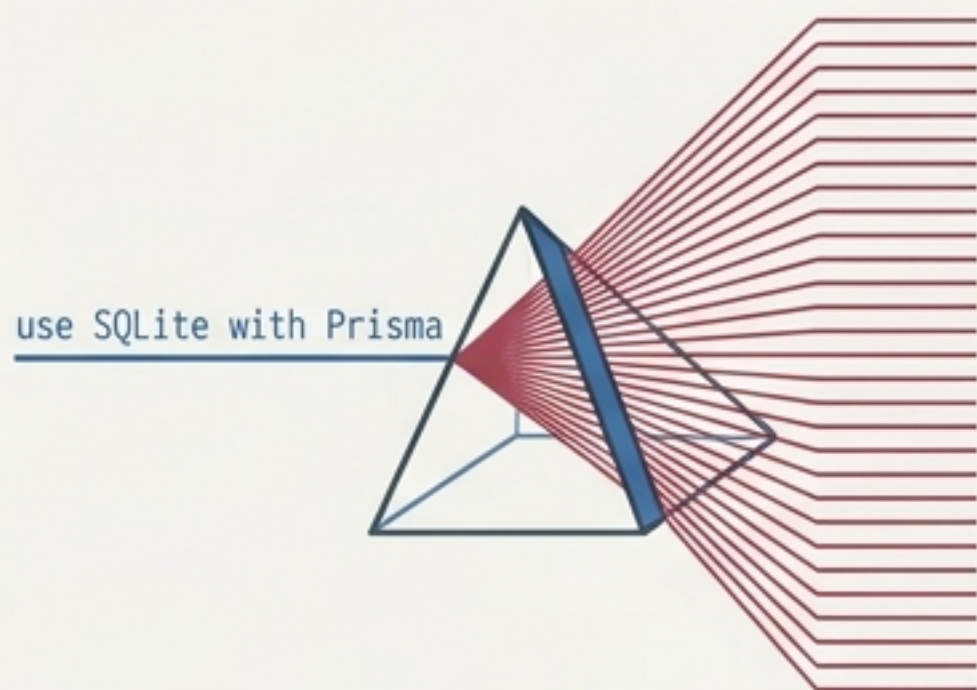
The Specification is a permanent technical artifact. It acts as an unbreakable contract not just between human teammates, but between human and agent.

Professional Software vs. Playing Around

	Vibe Coding	Spec-Driven Engineering
Output	Disposable "Slop" Code	Viable Software
Primary Artifact	Ephemeral Chat History	Version-Controlled Markdown Files
Scalability	Fails on large, ongoing codebases	Infinite scalability via Git branching
Cognitive Load	Severe Cognitive Debt (tracking evolving code)	Managed Load (reading high-level specs)
Primary Tool	Web Chatbots	Coding Agents integrated natively into IDE

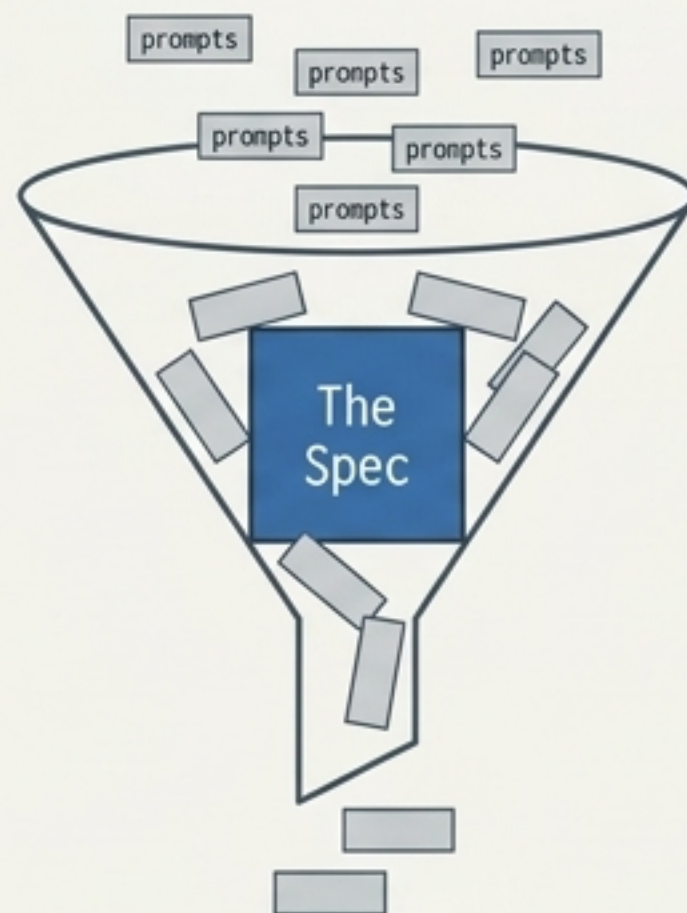
The Three Core Benefits of SDD

Downstream Amplification



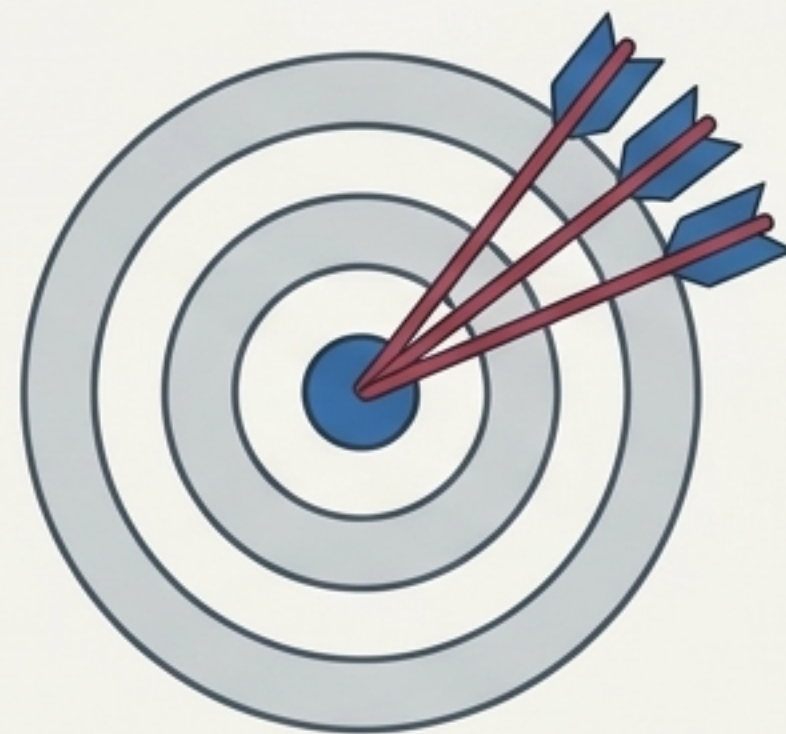
Small spec changes control massive code outputs. Changing one sentence to 'use MongoDB' instantly pivots the entire downstream generation.

Eliminating Context Decay



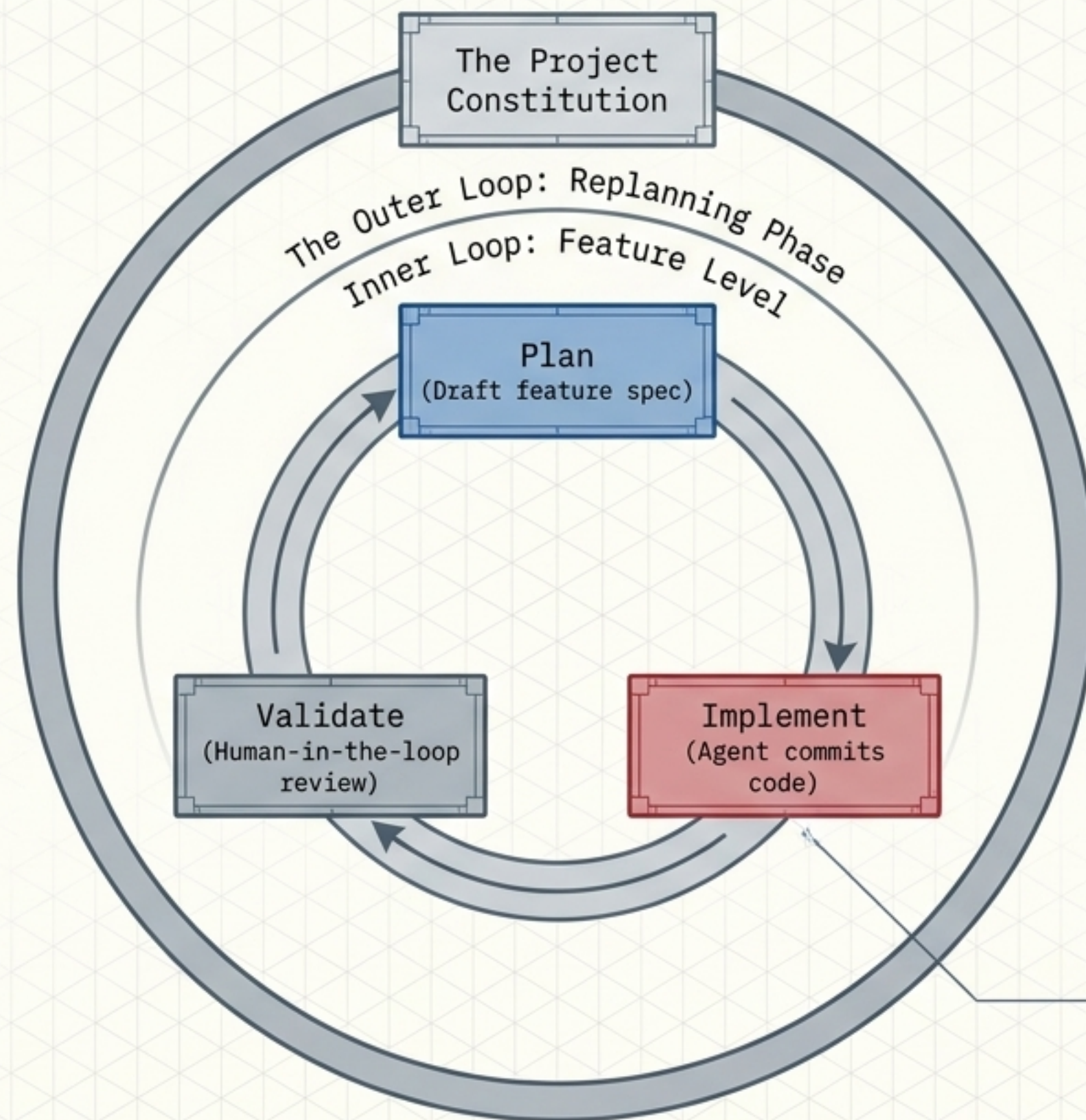
Persists project non-negotiables between multi-turn agent sessions. Anchors the agent's working memory.

Intent Fidelity



Forcing upfront constraints (success criteria, technical architecture) prevents the agent from making random, weird product choices.

The SDD Dual-Loop Workflow



Isolate each feature on its own **Git branch**. This ensures a clean slate between features, radically reducing context switching and AI hallucinations.

The Foundation: The Project Constitution

mission.md

Defines the Why.
Vision, Target
Audience, Tone.

Example: A playful Pet
Clinic parody app called
Agent Clinic.

tech-stack.md

Defines the Tools.

Example: Next.js backend,
React frontend, SQLite,
strict TypeScript.

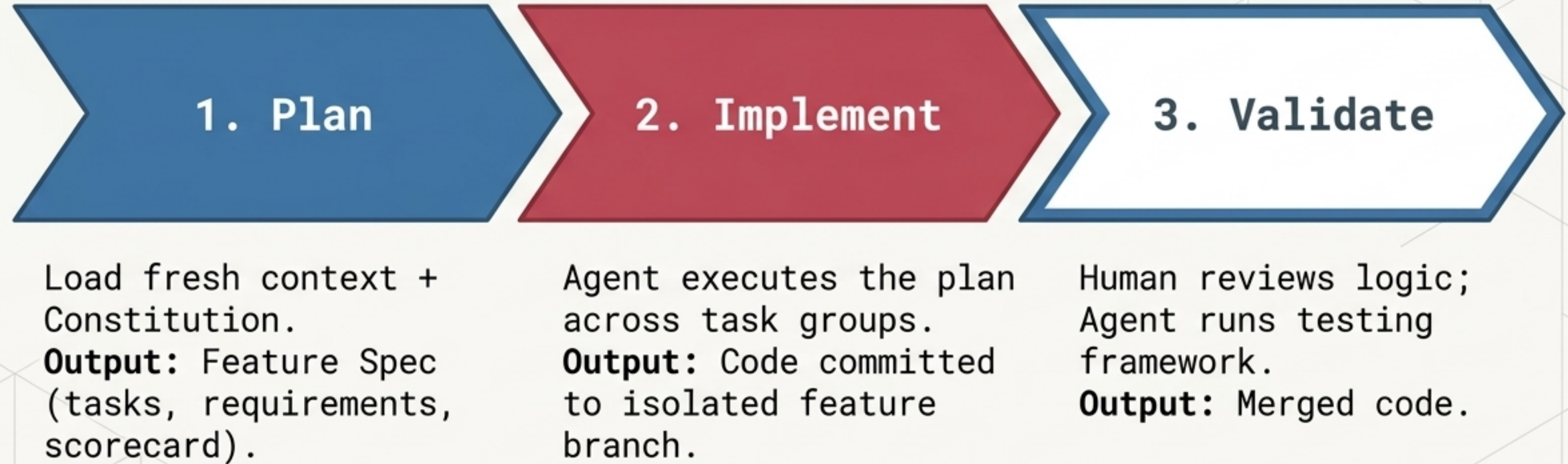
roadmap.mmd

Defines the Phases.
A living sequence of
manageable feature
branches.



The Constitution is drafted in a conversation with the agent. **The agent acts as an architect, asking you about trade-offs** (e.g., speed vs. data fidelity) you might not have considered.

The Inner Loop: Feature Development

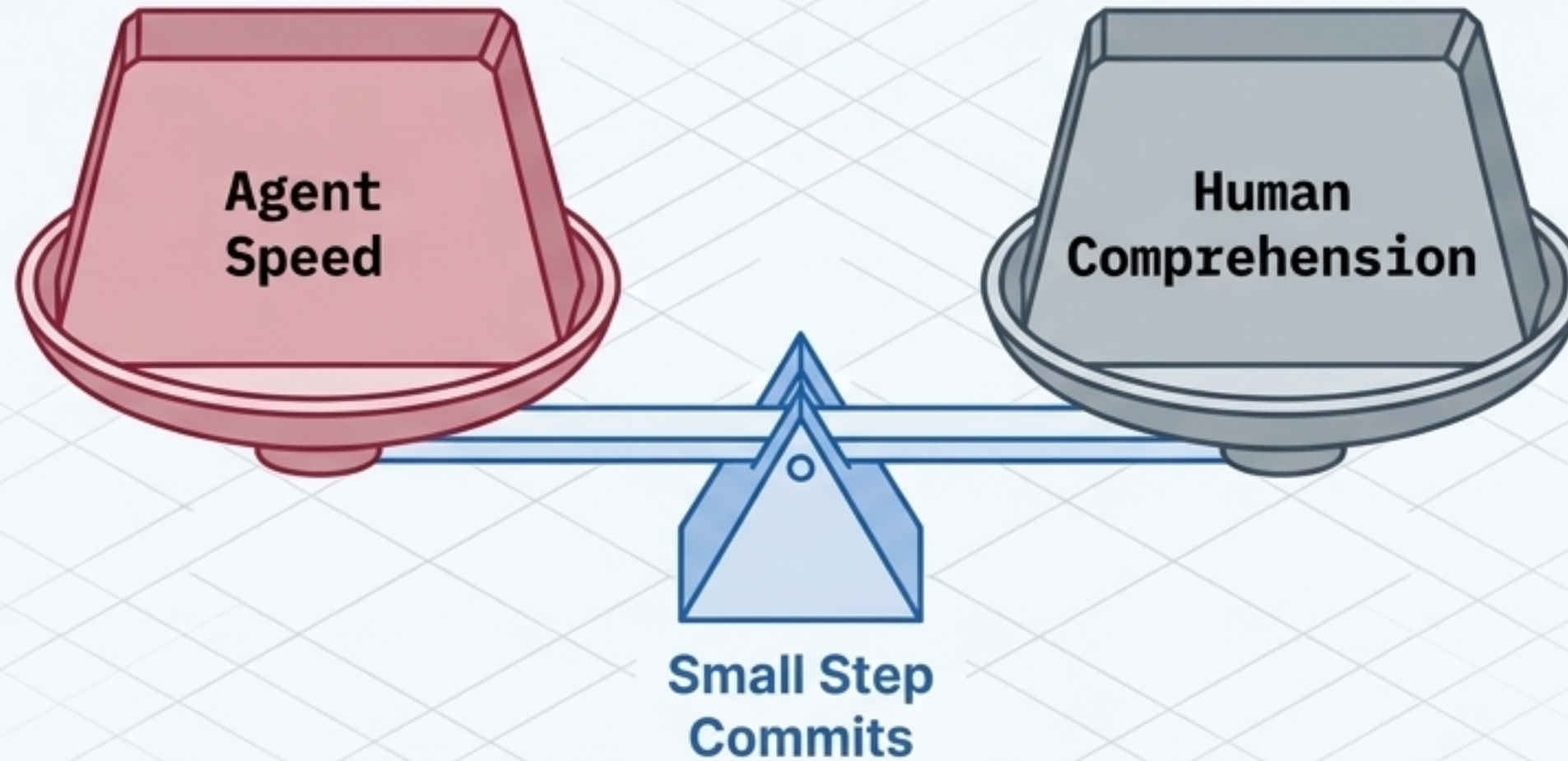


Never jump straight into implementation. The feature spec document is the critical intermediary step that prevents drift.

The Human-in-the-Loop Validation

The Problem

Generating massive code blocks instantly creates Cognitive Debt—the mental load required to track how code evolved without you.



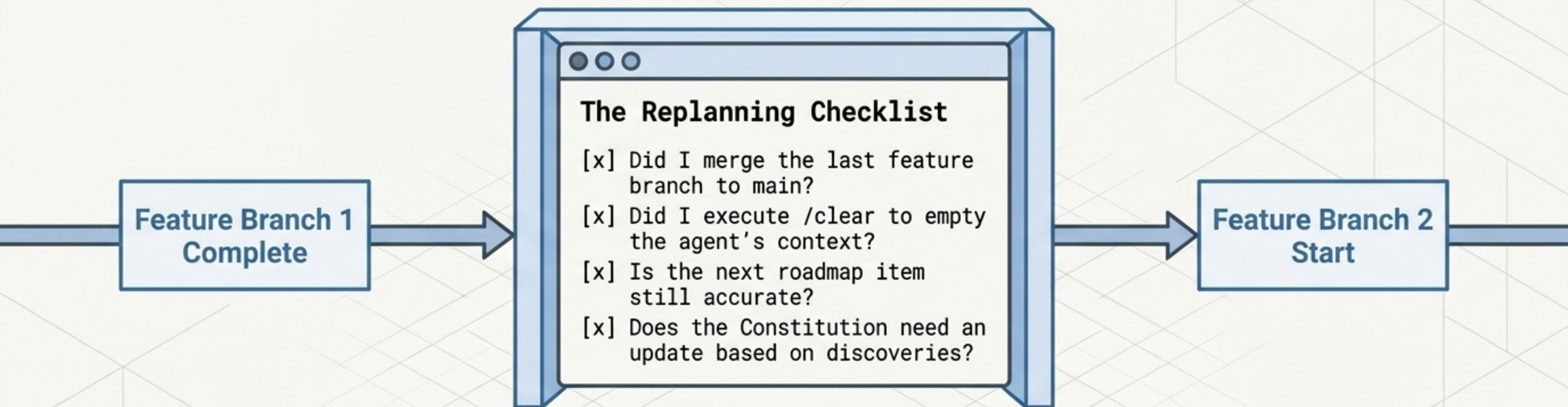
The Solution

Review for high-level intent fidelity, not low-level variables.

Verify layout architecture; ignore CSS class names.

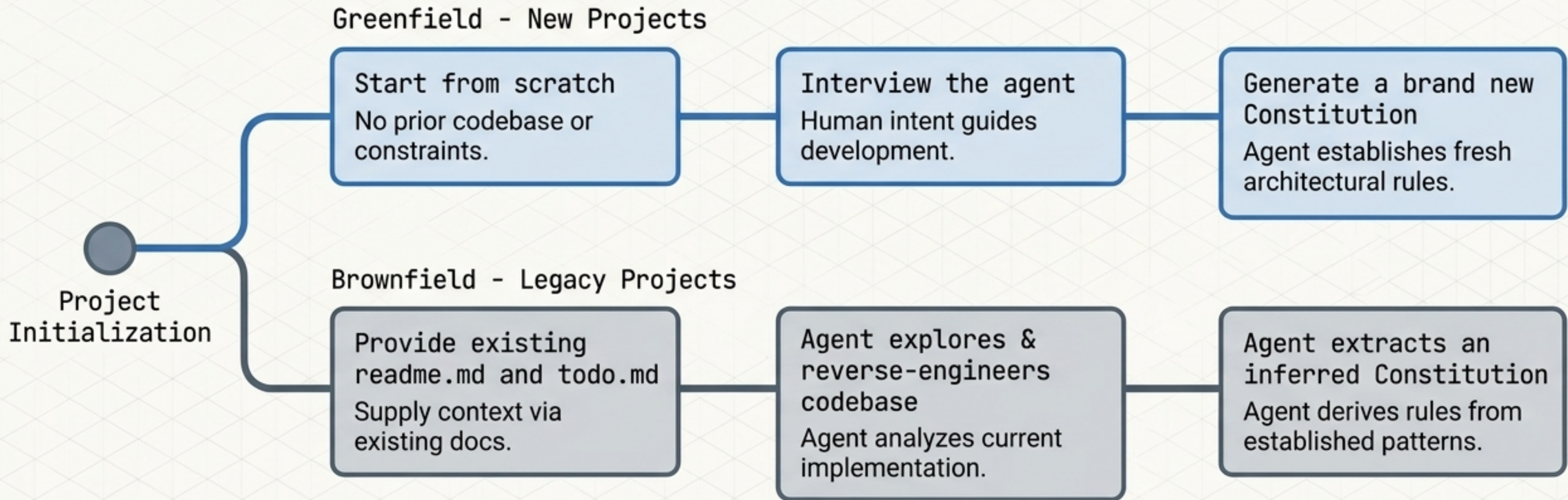
“Validation is not just checking for bugs. It is the mechanism by which you ensure you actually understand the code your agent just wrote.”

AI Fatigue & The Replanning Phase



You have to run slow to run fast. Never carry a massive, polluted memory snapshot into a new feature branch. Protect the agent's limited context budget.

Greenfield vs. Brownfield Projects



In legacy systems, the **Constitution** aligns the agent's future code changes with what past developers already created. It anchors the agent to established architectural rules.

Automating Workflow with Agent Skills

Write these three files, check the formatting, update the readme...



What are Skills?

A package of instructions and resources providing the agent with definable, repeatable workflows specific to your organization.

Examples in Action

Changelog Updater: Automatically updates stakeholders on every merge.

Validation Engine: Runs tests, lints, and formats in one command.

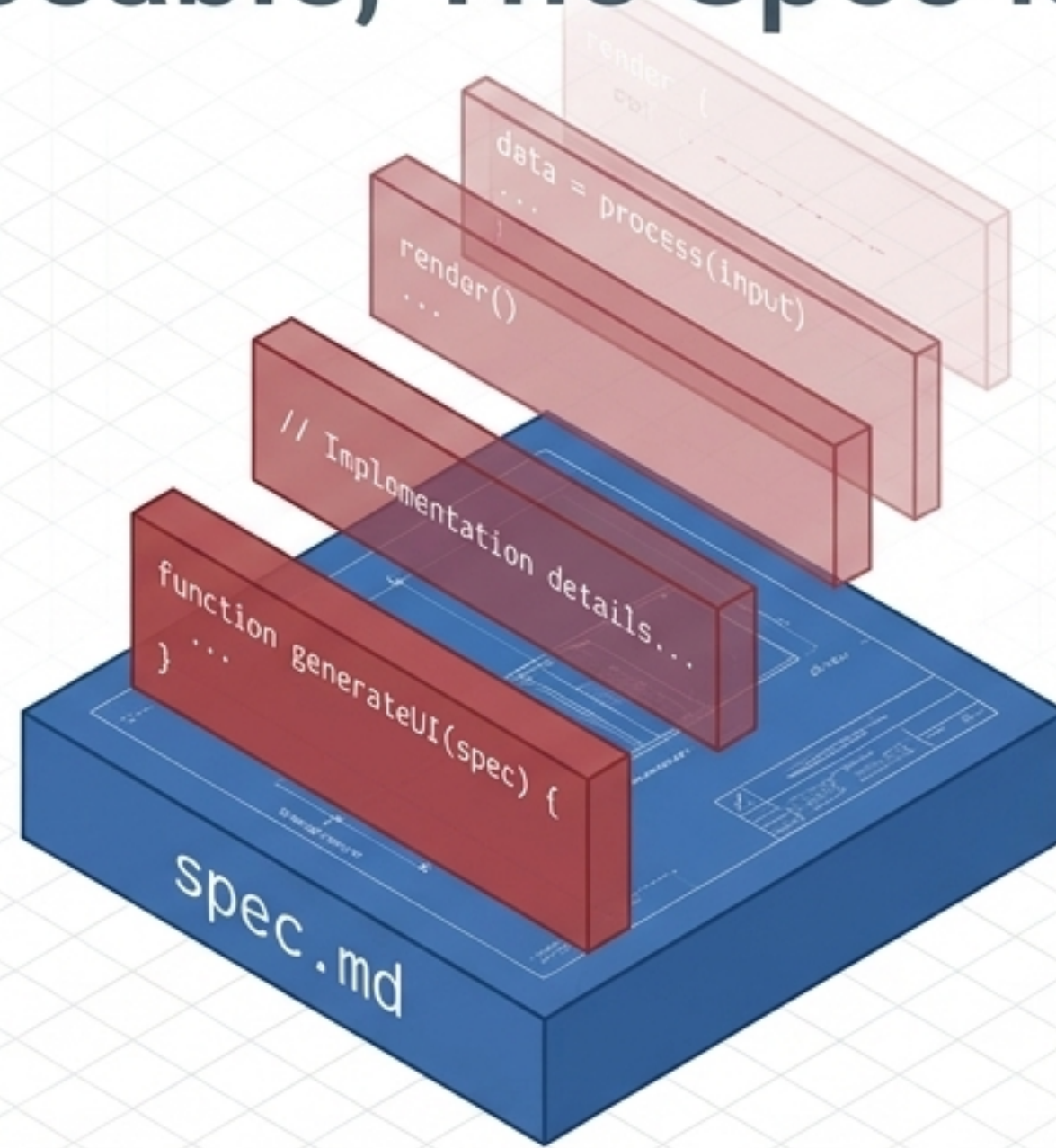
Pro Tip

You can use the agent's built-in skill creator to write its own customized skills.

Demystifying the Agent Ecosystem

Agent Skills	MCP (Model Context Protocol)	ACP (Agent Client Protocol)
Reusable prompt/workflow routines specific to your project.	Standard for connecting external data/APIs.	Standard for connecting agents to IDEs (similar to LSP).
Replaces manual chat typing. Packages organization-specific context.	Example: Context7 bringing updated React 9.2 documentation into the agent's context. Moving toward CLI + Skills for efficiency.	Enables plug-and-play integration. Example: Adding open-source agents directly into JetBrains.

Code is Disposable; The Spec is the Memory.



The ultimate shift in the developer's role. In the AI era, the codebase itself is ephemeral—it can be deleted and regenerated at will. The true, permanent artifact of value is the captured human intent. The spec is the memory of your project.

Bring Back Engineering.

Stop being a prompt-jockey hoping the AI gets it right.
Transition from building the How to architecting
the What and the Why.

**Remain the driver of your software.
The best code starts with a great spec.**